SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005

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CALIBRATION

Valid To: June 30, 2017 Certificate Number: 4041.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations[1]:

I. Dimensional

Parameter/Equipment	Range	CMC[2] (±)	Comments
Digital Caliper[3]			
External	Up to 500 mm	0.013 mm	Mitutoyo gage blocks and setting rods
Internal	Up to 500 mm	0.015 mm	
Depth Caliper[3]	Up to 150 mm	0.013 mm	Mitutoyo step gauge
Digital Micrometer[3]	Up to to 75 mm	0.003 mm	Mitutoyo gage blocks and Mitutoyo optical parallels
Dial Indicators[3]	Up to 20 µm (> 20 to 40) µm (>40 to 60) µm (>60 to 80) µm (>80 to 140) µm (>140 to 180) µm (>180 to 200) µm	0.25 µm 0.30 µm 0.35 µm 0.40 µm 0.45 µm 0.55 µm 0.65 µm	Mitutoyo make gage blocks

II. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
DC Voltage – Measure and Generate, Fixed Point[3]	1.018 V	2.5 µV	Fluke 732A
DC Voltage – Generate[3]			
Fixed Points	100 mV 1 V 10 V 100 V 1 kV	1.8 µV 7.5 µV 43 µV 0.53 mV 3.3 mV	Fluke 5700A
Ranged	100 mV to 1 V (>1 to 10) V (>10 to 100) V >100 V to 1 kV	8.9 µV 60 µV 0.86 mV 3.2 mV	
DC Current – Generate[3]			
Fixed Points	100 µA 1 mA 10 mA 100 mA 1 A	7.9 nA 72 nA 0.36 µA 4.0 µA 74 µA	Fluke 5700A
Ranged	100 µA to 1 mA (>1 to 10) mA (>10 to 100) mA >100 mA to 1 A	91 nA 0.72 µA 8.3 µA 0.14 mA	
	(10 to 300) A (>300 to 500) A	0.089 A 0.58 A	Fluke 5500A and coils

Parameter/Range	Frequency	CMC[2, 4] (±)	Comments
AC Voltage – Generate[3]			
10mV	40 Hz 1 kHz 20 kHz 100 kHz	9.1 µV 8.9 µV 9.0 µV 12 µV	Fluke 5700A
100mV	40 Hz 1 kHz 20 kHz 100 kHz 300 kHz	17 µV 16 µV 21 µV 43 µV 110 µV	
1V	40 Hz 1 kHz 20 kHz 100 kHz	0.1 mV 77 µV 0.14 mV 0.34 mV	
10V	40 Hz 1 kHz 20 kHz 100 kHz	0.65 mV 0.57 mV 1.2 mV 3.0 mV	
100V	40 Hz 1 kHz 20 kHz 100 kHz	7.5 mV 7.3 mV 11 mV 83 mV	
1kV	50 Hz 1 kHz	0.19 V 0.17 V	
(10 to 100) mV	40 Hz to 1 kHz >1 kHz to 20 kHz >20 kHz to 100 kHz	0.028 mV 0.049 mV 0.12 mV	
>100mV to 1V	40 Hz to 20 kHz >20 kHz to 100 kHz	0.22 mV 0.48 mV	
(>1 to 10) V	40 Hz to 1 kHz >1 kHz to 20 kHz >20 kHz to 100 kHz	2.1 mV 1.0 mV 4.3 mV	
(>10 to 100) V	40 Hz to 1 kHz >1 kHz to 20 kHz >20 kHz to 100 kHz	0.022 V 0.22 V 0.48 V	
>100 V to 1 kV	50 Hz to 1 kHz	0.21 V	

Parameter/Range	Frequency	CMC[2, 4] (±)	Comments
AC Current – Generate[3]			
1 mA 10 mA 100 mA 1 A	1 kHz 1 kHz 1 kHz 1 kHz	0.24 µA 2.5 µA 24 µA 0.48 mA	Fluke 5700A
(1 to 10) mA (>10 to 100) mA >100 mA to 1 A	1 kHz 1 kHz 1 kHz	3.1 µA 3.0 µA 0.87 mA	
(10 to 100) A (>100 to 300) A (>300 to 500) A	1 kHz 1 kHz 1 kHz	0.069 A 0.14 A 0.59 A	Fluke 5500A and coils

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Resistance – Generate[3]			
Fixed Points	10 Ω 100 Ω 1 kΩ 10 kΩ 100 kΩ 1 MΩ 10 MΩ	0.87 mΩ 4.9 mΩ 37 mΩ 0.33 Ω 3.9 Ω 57 Ω 1.3 kΩ	Fluke 5700A
Ranged	(0 to 19) Ω (>19 to 190) Ω >190Ω to 1.9 kΩ (>1.9 to 19) kΩ (>19 to 190) kΩ >190 kΩ to 1.9 MΩ (>1.9 to 290) MΩ	0.0018 Ω 0.13 Ω 0.7 Ω 8.8 Ω 130 Ω 0.7 kΩ 830 kΩ	Fluke 5500A
Capacitance – Generate[3]	400 pF to 20 nF (>20 to 100) nF (>100 to 300) nF >300 nF to 3 µF (>3 to 30) µF (>30 to 300) µF (>300 to 500) µF (>500 to 1000) µF	14 pF 0.065 nF 0.82 nF 6.0 nF 0.065 µF 0.72 µF 5.7 µF 20 µF	Fluke 5500A

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
DC Voltage – Measure[3]			
Fixed Points	100 mV	1.9 µV	HP-3458A
	1 V	20 µV	
	10 V	0.16 mV	
	100 V	0.59 mV	
	1 kV	7.1 mV	
Ranged	100 mV to 1 V	19 µV	
	(>1 to 10) V	0.16 mV	
	(>10 to 100) V	0.67 mV	
	>100 V to 1 kV	7.6 mV	
	(1 to 10) kV	9 V	Fluke 187 DMM and 80K-40 high voltage probe
	(>10 to 17) kV	9.6 V	
	(>17 to 25) kV	41 V	
DC Current – Measure[3]			
Fixed Points	100 µA	11 nA	HP 3458A
	1 mA	14 nA	
	10 mA	0.52 µA	
	100 mA	3.2 µA	
	1 A	64 µA	
Ranged	100 µA to 1 mA	21 nA	
	(>1 to 10) mA	0.4 µA	
	(>10 to 100) mA	4.6 µA	
	>100 mA to 1 A	0.14 mA	

Parameter/Range	Frequency	CMC[2, 4] (±)	Comments
AC Voltage – Measure[3]			
10 mV	40 Hz	0.14 mV	HP 3458A
	1 kHz	22 µV	
	20 kHz	11 µV	
	100 kHz	41 µV	
100 mV	40 Hz	19 µV	
	1 kHz	12 µV	
	20 kHz	23 µV	
	100 kHz	15 µV	
	300 kHz	15 µV	
1 V	40 Hz	90 µV	
	1 kHz	88 µV	
	20 kHz	0.17 mV	
	100 kHz	0.14 mV	

Parameter/Range	Frequency	CMC[2, 4] (±)	Comments
AC Voltage – Measure[3] (cont)			
10 V	40 Hz 1 kHz 20 kHz 100 kHz	8.4 mV 0.69 mV 1.5 mV 1.3 mV	HP-3458A
100 V	40 Hz 1 kHz 20 kHz 100 kHz	8.8 mV 7.9 mV 0.44 V 0.013 V	
10 mV to 100 mV	40 Hz to 20 kHz (>20 to 100) kHz	0.071 mV 0.096 mV	
>100 mV to 1 V	40 Hz to 20 kHz >20 kHz to 100 kHz	0.24 mV 0.96 mV	
(>1 to 10) V	40Hz to 20 kHz 20 kHz to 100 kHz	2.9 mV 9.7 mV	
(>10 to 100) V	40 Hz to 1 kHz (>1 to 100) kHz	0.03V 0.36V	
(1 to 6) kV (>6 to 10) kV	50 Hz 50 Hz	8.1V 21V	Fluke 187 DMM and 80K-40 high voltage probe
AC Current – Measure[3]			
1 mA 10 mA 100 mA 1 A	1 kHz 1 kHz 1 kHz 1 kHz	0.3 µA 0.37 µA 30 µA 0.44 mA	HP-3458A
(1 to 10) mA (>10 to 100) mA >100 mA to 1 A	1 kHz 1 kHz 1 kHz	20 µA 0.2 mA 2.1 mA	

Parameter/Equipment	Range	CMC[2, 4] (±)	Comments
Resistance – Measure[3]	10 Ω 100 Ω 1 kΩ 10 kΩ 100 kΩ 1 MΩ 10 MΩ	0.27 mΩ 1.3 mΩ 9.2 mΩ 0.19 Ω 7.1 Ω 25 Ω 0.68 kΩ	HP-3458A

III. Mechanical

Parameter/Equipment	Range	CMC[2] (±)	Comments
Torque Wrenches[3]	Up to 7.5 Nm (>7.5 to 10) Nm (>10 to 100) Nm (>100 to 300) Nm (>300 to 475) Nm	0.02 Nm 0.3 Nm 0.54 Nm 1 Nm 1.2 Nm	Imada I-80 for torque up to 7.5 Nm and Cedar DI-1M for torque up to 475 Nm
Scales & Balances[3]	Up to 100 g >100 g up to 200 g >200g up to 500g >500g up to 10 kg >10 kg up to 50 kg >50 kg up to 150 kg >150 kg up to 500 kg	0.6 mg 0.95 mg 1.3 mg 90 mg 6 g 8 g 60 g	Using appropriate combination of F1, F2 and M 2 Class weights
Pressure Gauge[3]	Up to 87.5 bar >87.5 up to 350 bar >350 up to 700 bar	0.12 bar 0.27 bar 0.44 bar	Fluke 700-RG31 reference pressure gage with Fluke pressure pump 700 H-TP2

IV. Optical Quantities

Parameter/Equipment	Range	CMC[2] (±)	Comments
Optical Power[3]			
For 850nm Wavelength	Up to 700 μW	21 μW	Using reference optical power meter EXFO: Model PM 1100
For 1310 nm Wave Length	Up to 10 μW >10 up to 100 μW > 100 up to 1000 μW	0.38 μW 0.79 μW 2.10 μW	Using reference optical power meter EXFO: Model PM 1100
For 1550 nm Wave Length	Up to 10 μW >10 up to 100 μW > 100 up to 1000 μW	0.38 μW 0.58 μW 7.30 μW	Using reference optical power meter EXFO: Model PM 1100

V. Thermodynamics

Parameter/Equipment	Range	CMC[2] (±)	Comments
Temperature – Measure, 4 Wire[3]	-80 °C to -18 °C >-18 °C to 300 °C >300 °C to 500 °C	0.18 °C 0.14 °C 2.7 °C	Pico PT-104 - 4 wire PT100

VI. Time & Frequency

Parameter/Equipment	Range	CMC[2] (±)	Comments
Frequency – Measure and Measuring Equipment[3]	10 Hz to 100 Hz >100 Hz to 1 GHz (1 to 22) GHz (>22 to 40) GHz	6 parts in 10^7 2.3 parts in 10^{10} 2.5 parts in 10^{10} 2.3 parts in 10^{10}	HP8340B, HP8648C, HP 5348A, FLUKE PM6685, HP8563E, Agilent N5183A, Agilent 8564EC, Datum LPRO 101 rubidium external time base

[1] This laboratory offers commercial calibration service.

[2] Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMC's represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of k = 2. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

[3] Field calibration service is available for this calibration and this laboratory meets A2LA *R104 – General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

[4] The stated measured values are determined using the indicated instrument (see Comments). This capability is suitable for the calibration of the devices intended to measure or generate the measured value in the ranges indicated. CMC's are expressed as either a specific value that covers the full range or as a percent or fraction of the reading plus a fixed floor specification.

Accredited Laboratory

A2LA has accredited

A.COM ELECTRONIC MEASUREMENT TECHNOLOGIES

Ra' anana, ISRAEL

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (*refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009*).

Presented this 15th day of April 2016.

President and CEO
For the Accreditation Council
Certificate Number 4041.01
Valid to June 30, 2017
Revised Janurary 18, 2017

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.